

Ryu Hemingway

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Software Engineering / AI Engineering Intern, Spring/Summer 2027

M.S. Computer Science candidate at Northeastern University, Silicon Valley (April 2028). Specializing in agentic AI workflows, LLM tool calling, RAG pipelines, and frontier and local model integration, shipped in production Python and Swift: a macOS AI assistant serving 20+ paying customers and an LLM order-intake and retrieval pipeline for a B2B wholesale distributor.

EDUCATION

Northeastern University, Silicon Valley, San Jose, CA September 2025 to April 2028 (expected)

Master of Science in Computer Science (Align Program) | GPA: 3.75 / 4.00

Coursework: Data Structures and Algorithms, Object-Oriented Design, Discrete Structures. In progress: Algorithms, Foundations of Artificial Intelligence

The Ohio State University, Columbus, OH | B.A. Psychology | Dean's List, Fall 2020

Aug 2017 to May 2021

St. George's University School of Medicine, Grenada | First year of M.D. program completed

Sep 2023 to Jun 2024

TECHNICAL SKILLS

Languages: Python, Swift, C, SQL, JavaScript, HTML/CSS

LLM and Agentic AI: LLM tool calling and function calling, agentic AI workflows, structured output with JSON schema validation, model routing and provider abstraction, prompt engineering, Anthropic Claude / Gemini / DeepSeek APIs, local inference (LM Studio, Ollama)

Retrieval and RAG: Retrieval-augmented generation, multi-format document ingestion, sentence-aware chunking, inverted indexes, BM25 ranking, SQLite FTS5 full-text search, incremental index sync, embeddings and semantic search, entity resolution

Evaluation and Reliability: Evaluation harnesses, held-out test sets, precision and recall, human-in-the-loop review queues, permission gating and risk tiering, data-residency constraints, CI-enforced test suites

Systems and Cloud: AWS (Lambda, DynamoDB, S3, CloudFront), serverless architecture, REST APIs, SQLite, ODBC, async/await concurrency, SwiftUI, AppKit, React, Stripe, Git / GitHub, GitHub Actions CI/CD, pytest, Swift Testing, Unix / Linux

EXPERIENCE

Leadway International, Hayward, CA September 2026 to Present

AI / Software Engineer, Part-Time Python, Anthropic Claude API, PostgreSQL, hybrid retrieval, structured output

- Scoped and won approval for an LLM order-intake system at a B2B wholesale distributor, converting orders that arrive as unstructured email and text into validated, schema-checked line items in place of manual re-entry.
- Architected the extraction pipeline as entity resolution over a catalog of near-identical SKUs: schema-validated structured output maps free-text line items to catalog SKUs and normalizes case, pack, and each quantities, routing low-confidence lines to a human-in-the-loop review queue with drafted clarifying questions instead of guessing.
- Designed a data path with no write access to the ERP: the item master arrives as a vendor-run export rather than a live database connection, and output is a structured file for human import. Made the inference endpoint a config value so inference can move to AWS Bedrock or Google Vertex AI inside the client tenant when data residency requires it.
- Designed the retrieval layer as hybrid search over the product catalog: Postgres full-text and vector similarity fused and reranked, with hard attribute filters so a near-miss on flavor or pack type is never a candidate, and per-account ordering history mined from past orders as a disambiguation signal.
- Defined the evaluation plan before the build: extraction accuracy and review-queue rate measured against a held-out set of historical orders, with a pilot accuracy threshold that gates rollout.

Integrated Agentic, San Jose, CA

January 2026 to Present

Founder and Sole Engineer | Ghost, native macOS AI assistant Swift, SwiftUI, AWS, SQLite

- Shipped a native macOS AI assistant as sole engineer, owning system design, code review, and release. A provider-abstraction and model-routing layer fronts eight local and hosted LLM providers behind a deterministic intent classifier.
- Designed an 80-tool function-calling harness with four risk tiers, permission gating, and an undo journal as runtime guardrails. Added a three-state outcome model (failed, verified, claimed-without-proof) after finding the agent logged success for actions it had not completed.
- Built a fully local RAG pipeline over 30+ file formats: sentence-aware 3,500-character chunking with 500-character overlap, BM25 ranking on an SQLite FTS5 inverted index, and incremental index sync through FSEvents, moved off the main thread. A privacy invariant keeps Messages, Notes, Mail, and Contacts out of every cloud request.
- Instrumented the project with 2,141 Swift Testing cases that run in CI on every push, plus a metrics script that regenerates every published figure directly from the working tree.
- Shipped signed, notarized releases to 20+ paying customers on a Stripe-backed AWS Lambda and DynamoDB licensing service. Every release since 2.0 traces to a user-reported issue.

UCSF Benioff Children's Hospital Oakland, Oakland, CA

August 2021 to May 2022

Medical Scribe, Pediatric Orthopedics | Certified Medical Scribe, ScribeAmerica

- Documented 800+ hours of diagnostic data across concurrent patient cases in a high-volume EMR system, and built documentation protocols the scribe team adopted.

PROJECTS

Huffman Coding: Lossless Data Compression Research | Python 2026

- Research paper on the exchange-argument optimality proof, a from-scratch implementation with 22 unit tests, and seeded benchmarks measuring 0.158 μ s to encode and 0.463 μ s to decode per character, matching the $\mathcal{O}(m)$ prediction.

Programming Fundamentals and AI: Terminal Curriculum | Python, C, Java

2026

- Offline-first terminal learning app of 244 lessons. Grades submissions by compiling and running the code (gcc, javac) and returns the actual compiler error plus an output diff.

LEADERSHIP AND VOLUNTEER

- American Red Cross, Disaster Action Team volunteer (2019 to 2022): emergency response logistics and case tracking for field teams.
- Operation Smile, Event Organizer, Bangkok (2015): raised \$30,000+ for cleft palate surgeries.